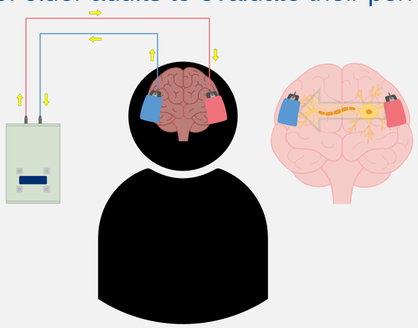
Comparison of GRACE and SynthSeg Deep Learning Models for Whole-Head Brain Segmentation

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Introduction

- **Transcranial Direct Current Stimulation (tDCS)** shows potential for **treating neurological diseases**, but **anatomical differences** among patients affect consistency.
- **Whole-head brain segmentation** helps optimize tDCS but is **time-consuming**, taking up to **30 hours per patient**.
- Deep learning models, such as **GRACE** and **SynthSeg**, provide fully **automated segmentation**, and this study trained these models using **MRI scans** of older adults to evaluate their performance.

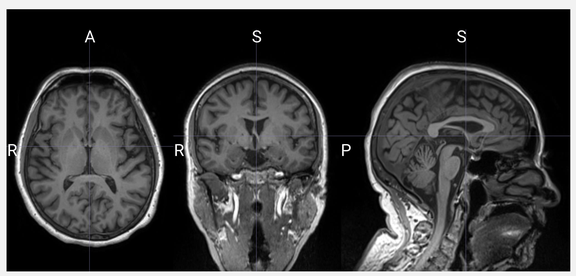


Transcranial Direct Current Stimulation Setup

Dataset

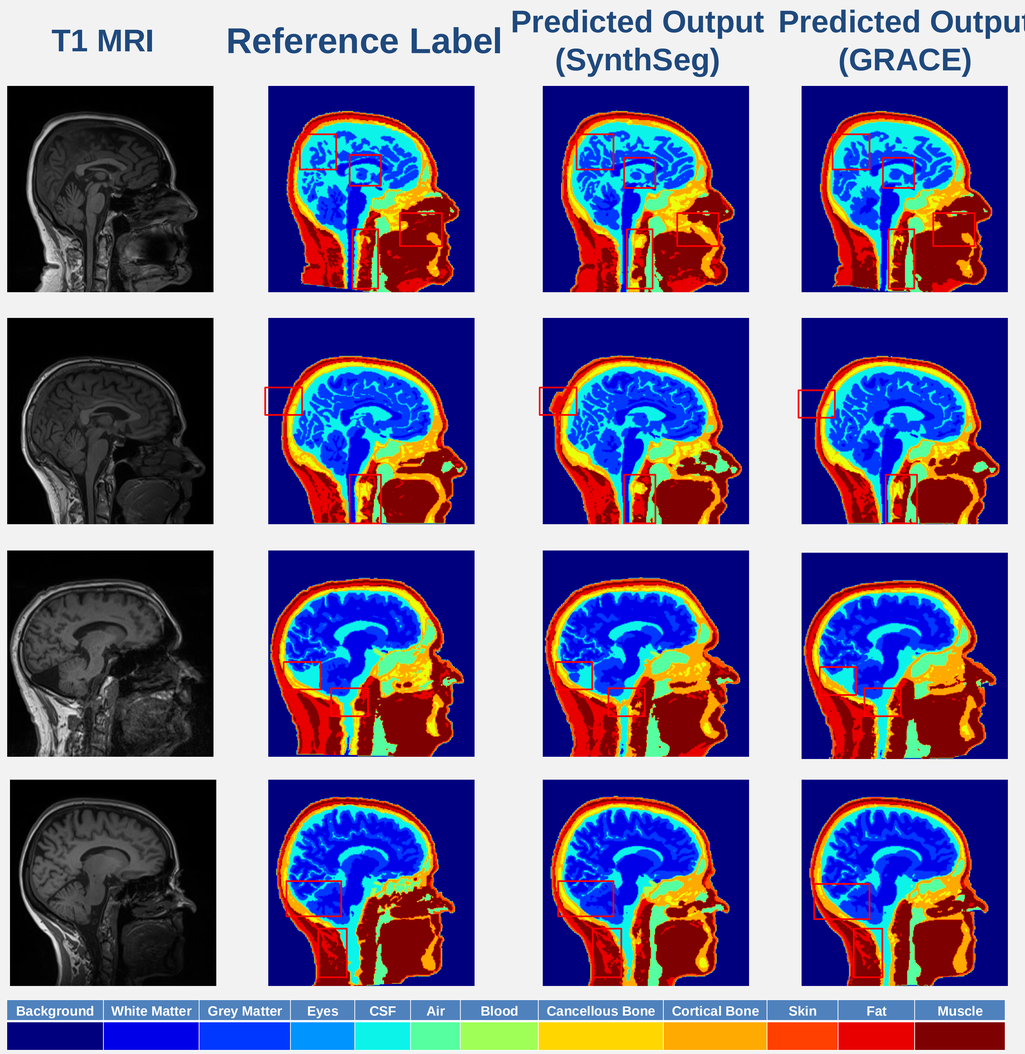
- Dataset

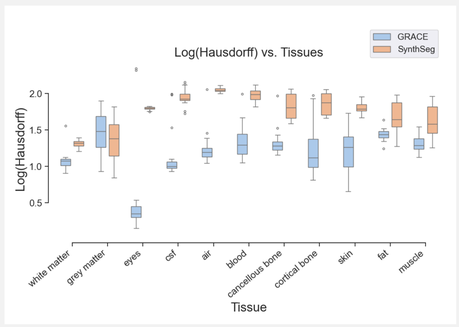
- This investigation used a dataset from the **Augmenting Cognitive Training in Older Adults** (ACT) trial, which combined cognitive training with tDCS to improve cognitive function.
- The subset included 177 MRI images (176x256x256 voxels) of cognitively healthy adults aged 65-89.



Example of ACT T1 MRI Scan

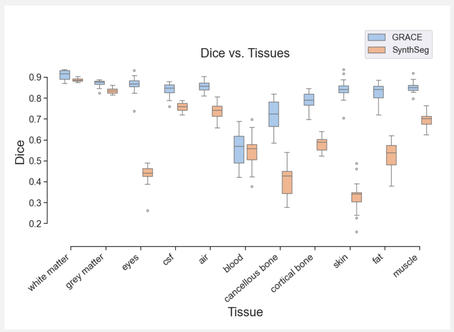
SynthSeg and Grace Segmentation Results





Hausdorff Distance per Tissue Type

(Lower Scores Indicate Better Performance)



Dice Score per Tissue Type

(Higher Scores Indicate Better Performance)

Methods

- **Experiment details**
  - Each image was segmented into **11 tissue labels**: white matter, grey matter, eyes, cerebrospinal fluid (CSF), air, major artery (blood), cancellous bone, cortical bone, skin, fat, and muscle, using a semi-automatic process by trained professionals.
  - The dataset was split into 157 training volumes and 20 testing volumes, and both the SynthSeg and GRACE models were **trained** using their recommended parameters.
  - The trained models were then used to **segment** the 20 testing volumes.
  - Predicted segmentations were compared against reference annotations.
  - Segmentation results from the testing set were evaluated using metrics like the DICE score and Hausdorff distance.

Results

- GRACE achieves a **lower average Hausdorff** distance compared to SynthSeg.
- GRACE shows a **higher average DICE** score compared to SynthSeg.
- Lower Hausdorff distances and higher DICE scores indicate greater model accuracy.
- GRACE is the **most accurate at segmenting** the tissues to optimize tDCS simulation applications.
- SynthSeg is a self-supervised model trained on unlabeled data, so it is **more generalizable**.

Citations

1. S. E. Stolte et al., “Precise and rapid whole-head segmentation from magnetic resonance images of older adults using Deep Learning,” Imaging Neuroscience (Cambridge, Mass.)
2. B. Billot et al., “Synthseg: Segmentation of brain MRI scans of any contrast and resolution without retraining,” ScienceDirect
3. A. Woods, R. Cohen, M. Marsiske, and G. Alexander, “Augmenting cognitive training in older adults (the ACT study): Design and methods of a phase III tDCS and cognitive training trial,” Contemporary Clinical Trials

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